ROCSCIENCE

SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

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Slide2 Analysis Information

SLIDE - An Interactive Slope Stability Program

Project Summary

Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:01.250s
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	30
Tolerance:	1e-05
Maximum number of iterations:	50
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Ru Coefficient
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No
Seismic Load Coefficient (Horizontal):	0.132

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	20 kN/m3
Cohesion	25 kPa
Phi	30 °
Ru Value	0.5

Global Minimums

Method: bishop simplified

FS	0.987678
Center:	20.928, 88.470
Radius:	90.914
Left Slip Surface Endpoint:	-0.009, 0.000
Right Slip Surface Endpoint:	86.020, 25.000
Resisting Moment:	594486 kN-m
Driving Moment:	601902 kN-m
Total Slice Area:	851.266 m2
Surface Horizontal Width:	86.0284 m
Surface Average Height:	9.89518 m

Method: spencer

FS	1.001120
Center:	20.735, 88.244
Radius:	90.649
Left Slip Surface Endpoint:	-0.009, 0.000
Right Slip Surface Endpoint:	85.677, 25.000
Resisting Moment:	594924 kN-m
Driving Moment:	594261 kN-m
Resisting Horizontal Force:	6192.91 kN
Driving Horizontal Force:	6186.01 kN
Total Slice Area:	841.124 m2
Surface Horizontal Width:	85.6859 m
Surface Average Height:	9.81637 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	13355
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	12359
Number of Invalid Surfaces:	996

Error Codes

Error Code -108 reported for 294 surfaces
Error Code -111 reported for 148 surfaces
Error Code -112 reported for 554 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge
-112 = The coefficient M-Alpha = cos(alpha)(1+tan(alpha)tan(phi)/F) < 0.2 for the final iteration of the safety factor calculation. This screens out some slip surfaces which may not be valid in the context of the analysis, in particular, deep seated slip surfaces with many high negative base angle slices in the passive zone.

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.987678

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.86761	45.3082	-12.3893	Material 1	25	30	34.3393	33.9161	23.3432	7.89998	15.4432	15.8	7.89998
2	2.86761	133.5	-10.5449	Material 1	25	30	43.6707	43.1326	54.6839	23.2773	31.4066	46.5546	23.2773
3	2.86761	216.229	-8.71151	Material 1	25	30	52.0092	51.3683	83.3731	37.702	45.6711	75.4039	37.7019
4	2.86761	293.583	-6.8871	Material 1	25	30	59.431	58.6987	109.557	51.1895	58.3677	102.379	51.1894
5	2.86761	365.632	-5.06968	Material 1	25	30	66.001	65.1877	133.359	63.752	69.6071	127.504	63.7519
6	2.86761	432.429	-3.25737	Material 1	25	30	71.7743	70.8899	154.882	75.3987	79.4836	150.797	75.3987
7	2.86761	494.009	-1.44832	Material 1	25	30	76.798	75.8517	174.214	86.136	88.0777	172.272	86.136
8	2.86761	550.394	0.35929	Material 1	25	30	81.1126	80.1131	191.426	95.9673	95.4586	191.935	95.9673
9	2.86761	601.587	2.16725	Material 1	25	30	84.7528	83.7085	206.58	104.893	101.687	209.787	104.894
10	2.86761	647.579	3.97738	Material 1	25	30	87.7488	86.6676	219.724	112.913	106.811	225.825	112.912
11	2.86761	688.342	5.7915	Material 1	25	30	90.1265	89.016	230.899	120.02	110.879	240.04	120.02
12	2.86761	723.834	7.61147	Material 1	25	30	91.9081	90.7756	240.135	126.209	113.926	252.417	126.208
13	2.86761	753.996	9.43921	Material 1	25	30	93.1126	91.9653	247.455	131.468	115.987	262.935	131.467
14	2.86761	778.749	11.2767	Material 1	25	30	93.7565	92.6012	252.872	135.784	117.088	271.567	135.783
15	2.86761	797.998	13.126	Material 1	25	30	93.8534	92.6969	256.395	139.14	117.255	278.28	139.14
16	2.86761	811.626	14.9894	Material 1	25	30	93.4148	92.2637	258.02	141.516	116.504	283.032	141.516
17	2.86761	819.494	16.8692	Material 1	25	30	92.4501	91.3109	257.742	142.888	114.854	285.776	142.888
18	2.86761	821.437	18.7679	Material 1	25	30	90.9666	89.8457	255.543	143.227	112.316	286.454	143.227
19	2.86761	817.262	20.6882	Material 1	25	30	88.9702	87.8739	251.399	142.499	108.9	284.997	142.498
20	2.86761	806.744	22.6332	Material 1	25	30	86.4645	85.3991	245.279	140.665	104.614	281.33	140.665
21	2.86761	789.62	24.6062	Material 1	25	30	83.4521	82.4238	237.14	137.679	99.4608	275.358	137.679
22	2.86761	765.584	26.6108	Material 1	25	30	79.9334	78.9485	226.93	133.488	93.4416	266.976	133.488
23	2.86761	734.277	28.6513	Material 1	25	30	75.9078	74.9725	214.584	128.029	86.5553	256.059	128.03
24	2.86761	695.28	30.7324	Material 1	25	30	71.3731	70.4936	200.027	121.23	78.797	242.46	121.23
25	2.86761	648.097	32.8594	Material 1	25	30	66.3253	65.508	183.165	113.003	70.1617	226.006	113.003
26	2.86761	592.14	35.0389	Material 1	25	30	60.7595	60.0108	163.887	103.246	60.6408	206.493	103.247
27	2.86761	507.234	37.2783	Material 1	25	30	53.2962	52.6395	136.315	88.4419	47.8728	176.884	88.4418
28	2.86761	377.321	39.5866	Material 1	25	30	42.9902	42.4605	96.0325	65.7901	30.2424	131.58	65.79
29	2.86761	235.348	41.9747	Material 1	25	30	32.3091	31.911	53.0058	41.0356	11.9702	82.0712	41.0356
30	2.86761	80.686	44.4563	Material 1	25	30	21.3119	21.0493	7.22576	14.0685	-6.84274	28.137	14.0685

Global Minimum Query (spencer) - Safety Factor: 1.00112

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.8562	44.8213	-12.3047	Material 1	25	30	46.5189	46.571	45.2084	7.84633	37.3621	35.0617	27.2153
2	2.8562	132.066	-10.4628	Material 1	25	30	58.338	58.4033	80.9753	23.1191	57.8562	70.2022	47.0831
3	2.8562	213.9	-8.63182	Material 1	25	30	67.9588	68.0349	111.984	37.4449	74.5389	101.667	64.2224
4	2.8562	290.412	-6.8097	Material 1	25	30	75.7365	75.8213	138.864	50.8389	88.0248	129.82	78.9808
5	2.8562	361.668	-4.99447	Material 1	25	30	81.9485	82.0403	162.11	63.3129	98.7968	154.948	91.6352
6	2.8562	427.722	-3.18427	Material 1	25	30	86.8156	86.9128	182.112	74.8761	107.236	177.283	102.406
7	2.8562	488.607	-1.37724	Material 1	25	30	90.5152	90.6166	199.186	85.5346	113.651	197.01	111.475
8	2.8562	544.344	0.428413	Material 1	25	30	93.1927	93.2971	213.586	95.2918	118.294	214.283	118.991
9	2.8562	594.937	2.23449	Material 1	25	30	94.9677	95.0741	225.52	104.148	121.372	229.226	125.078
10	2.8562	640.374	4.0428	Material 1	25	30	95.9406	96.0481	235.162	112.103	123.059	241.942	129.839
11	2.8562	680.628	5.85516	Material 1	25	30	96.1961	96.3038	242.651	119.149	123.502	252.516	133.367
12	2.8562	715.656	7.67342	Material 1	25	30	95.8064	95.9137	248.108	125.281	122.827	261.016	135.735
13	2.8562	745.4	9.4995	Material 1	25	30	94.8339	94.9401	251.628	130.488	121.14	267.497	137.009
14	2.8562	769.782	11.3354	Material 1	25	30	93.333	93.4375	253.294	134.756	118.538	272.003	137.247
15	2.8562	788.705	13.1831	Material 1	25	30	91.3517	91.454	253.171	138.069	115.102	274.569	136.5
16	2.8562	802.054	15.045	Material 1	25	30	88.9326	89.0322	251.313	140.406	110.907	275.217	134.811
17	2.8562	809.691	16.9233	Material 1	25	30	86.1145	86.2109	247.763	141.743	106.02	273.965	132.222
18	2.8562	811.451	18.8205	Material 1	25	30	82.9324	83.0253	242.554	142.051	100.503	270.819	128.768
19	2.8562	807.142	20.7394	Material 1	25	30	79.4195	79.5084	235.708	141.297	94.4111	265.781	124.484
20	2.8562	796.541	22.683	Material 1	25	30	75.6066	75.6913	227.241	139.441	87.7998	258.841	119.4
21	2.8562	779.387	24.6546	Material 1	25	30	71.5238	71.6039	217.158	136.438	80.7201	249.987	113.549
22	2.8562	755.374	26.6579	Material 1	25	30	67.2006	67.2759	205.458	132.234	73.2243	239.195	106.961
23	2.8562	724.147	28.6971	Material 1	25	30	62.6669	62.7371	192.13	126.768	65.3625	226.435	99.6674
24	2.8562	685.29	30.7768	Material 1	25	30	57.9538	58.0187	177.155	119.965	57.1905	211.671	91.7061
25	2.8562	638.309	32.9027	Material 1	25	30	53.0943	53.1538	160.505	111.741	48.764	194.857	83.1157
26	2.8562	582.619	35.0809	Material 1	25	30	48.125	48.1789	142.139	101.992	40.1472	175.938	73.9461
27	2.8562	502.701	37.3191	Material 1	25	30	42.4573	42.5049	118.321	88.0019	30.3194	150.688	62.6856
28	2.8562	374.825	39.6262	Material 1	25	30	35.381	35.4206	83.665	65.6161	18.0489	112.962	47.3458
29	2.8562	233.786	42.0132	Material 1	25	30	28.6168	28.6488	47.2461	40.9262	6.31992	73.0247	32.0985
30	2.8562	80.1494	44.4937	Material 1	25	30	22.3318	22.3568	9.45264	14.0308	-4.57816	31.3932	17.3624

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.987678

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	-0.00870632	0	0	0	0
2	2.85891	-0.629922	107.195	0	0
3	5.72652	-1.16373	243.994	0	0
4	8.59413	-1.60312	401.227	0	0
5	11.4617	-1.94949	570.846	0	0
6	14.3294	-2.20388	745.774	0	0
7	17.197	-2.36709	919.791	0	0
8	20.0646	-2.43959	1087.44	0	0
9	22.9322	-2.42161	1243.94	0	0
10	25.7998	-2.31309	1385.15	0	0
11	28.6674	-2.1137	1507.49	0	0
12	31.535	-1.82285	1607.92	0	0
13	34.4026	-1.43964	1683.91	0	0
14	37.2702	-0.962898	1733.42	0	0
15	40.1379	-0.391105	1754.89	0	0
16	43.0055	0.277583	1747.24	0	0
17	45.8731	1.04539	1709.88	0	0
18	48.7407	1.91495	1642.69	0	0
19	51.6083	2.88937	1546.11	0	0
20	54.4759	3.97228	1421.12	0	0
21	57.3435	5.1679	1269.32	0	0
22	60.2111	6.48117	1092.97	0	0
23	63.0788	7.91784	895.105	0	0
24	65.9464	9.48465	679.642	0	0
25	68.814	11.1895	451.518	0	0
26	71.6816	13.0418	216.895	0	0
27	74.5492	15.0526	-16.5813	0	0
28	77.4168	17.2354	-228.254	0	0
29	80.2844	19.6066	-382.491	0	0
30	83.152	22.1863	-457.647	0	0
31	86.0196	25	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.00112

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	-0.00870632	0	0	0	0
2	2.84749	-0.622995	155.117	55.3294	19.631
3	5.70369	-1.15044	347.021	123.781	19.6311
4	8.55988	-1.58402	561.446	200.265	19.6311
5	11.4161	-1.92509	786.794	280.646	19.6311
6	14.2723	-2.1747	1013.58	361.54	19.6311
7	17.1285	-2.3336	1234.02	440.171	19.6312
8	19.9847	-2.40227	1441.74	514.261	19.631
9	22.8409	-2.38091	1631.5	581.949	19.6311
10	25.6971	-2.26947	1799.09	641.726	19.6311
11	28.5533	-2.0676	1941.11	692.386	19.6311
12	31.4095	-1.7747	2054.96	732.993	19.631
13	34.2657	-1.38987	2138.66	762.849	19.6311
14	37.1218	-0.911937	2190.87	781.472	19.6311
15	39.978	-0.339378	2210.81	788.586	19.6311
16	42.8342	0.32965	2198.24	784.103	19.6311
17	45.6904	1.09737	2153.45	768.124	19.631
18	48.5466	1.96642	2077.21	740.932	19.6311
19	51.4028	2.93988	1970.86	702.995	19.631
20	54.259	4.0214	1836.23	654.975	19.6311
21	57.1152	5.21517	1675.76	597.737	19.6311
22	59.9714	6.52614	1492.49	532.363	19.631
23	62.8276	7.96003	1290.11	460.177	19.6311
24	65.6838	9.52356	1073.11	382.774	19.6311
25	68.54	11.2246	846.83	302.06	19.6311
26	71.3962	13.0726	617.618	220.302	19.6311
27	74.2524	15.0785	393.045	140.197	19.631
28	77.1086	17.2559	190.33	67.8897	19.631
29	79.9648	19.6209	44.0364	15.7076	19.6311
30	82.821	22.1938	-26.648	-9.50522	19.6311
31	85.6772	25	0	0	0

Discharge Sections

Entity Information

External Boundary

X	Y
-50	0
-50	-25
150	-25
150	25
75	25
0	0